Functional and genomic analysis of native Colombian yeast strains with potential use as starters in the production of specialty coffee

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Abstract

Coffee is one of the leading agricultural products in Colombia and one of the most widely consumed beverages worldwide. This drives the coffee industry to implement strategies that enhance the coffee quality. It has been established that the fermentation process is a crucial factor influencing sensorial coffee attributes. During the process, volatile organic compounds (VOCs) produced by yeasts' metabolism play a decisive role in determining the final quality. Previous studies carried out in order to characterize the metabolic profiles of the microorganisms involved in the fermentation identified three native yeasts related to the production of VOCs that contribute to the desired sensory characteristics in coffee. These strains have biotechnological potential to be used as starter cultures to assist the fermentation process in Colombia. Yeast strains were isolated, and their genomes were sequenced using the Illumina NovaSeq sequencing platform. Sequence analysis revealed that the isolates are closely related to *Starmerella bacillaris*, *Hanseniaspora uvarum* and *Starmerella sorbosivorans*. Analysis of the encoded proteins revealed genes involved in the assimilation of coffee mucilage and the synthesis of secondary metabolites associated with the improved qualities during fermentation. This information provides a genetic framework for the future use of these strains as starter cultures in producing high-quality cofees.

Methodology

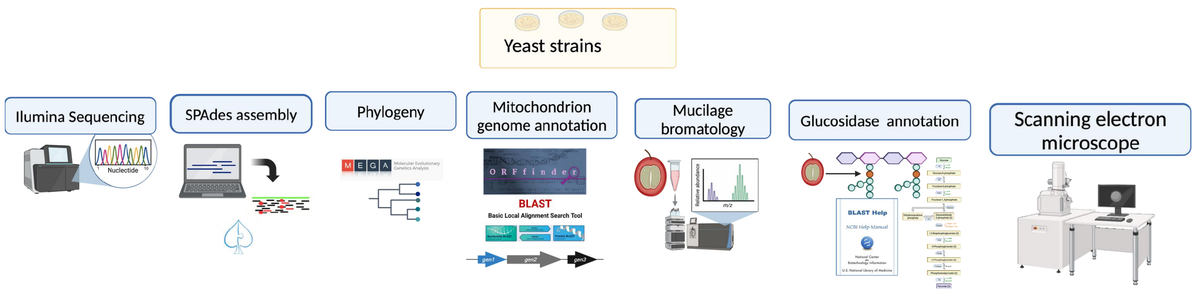

Figure 1. Methodology

Results

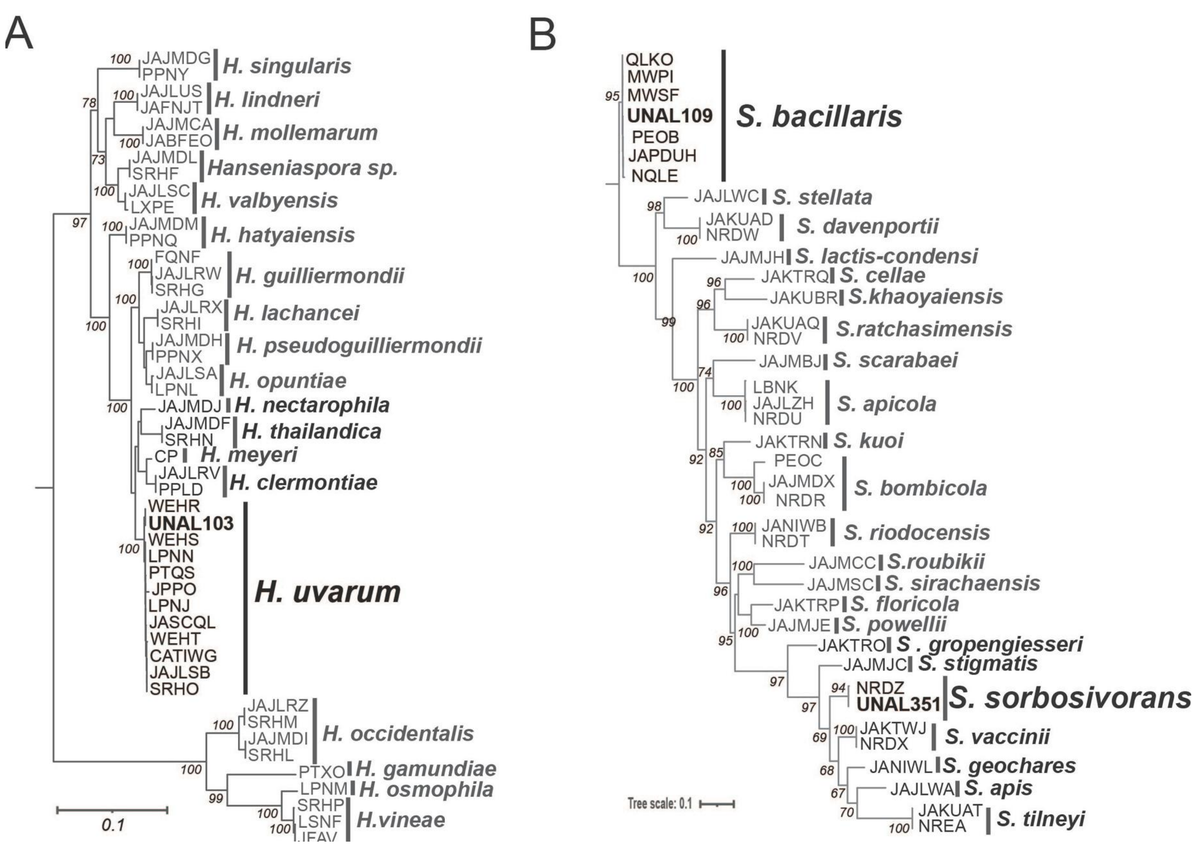

Figure 2. Phylogenetic trees

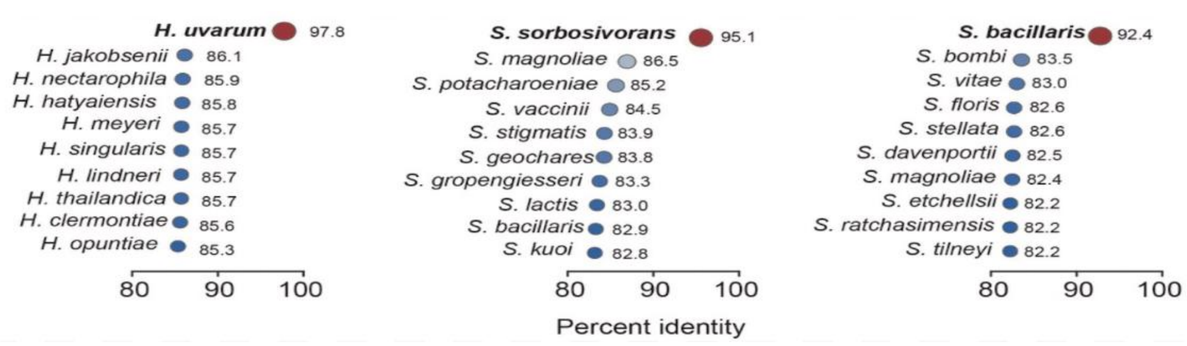

Figure 3. Genome Similarity Comparison

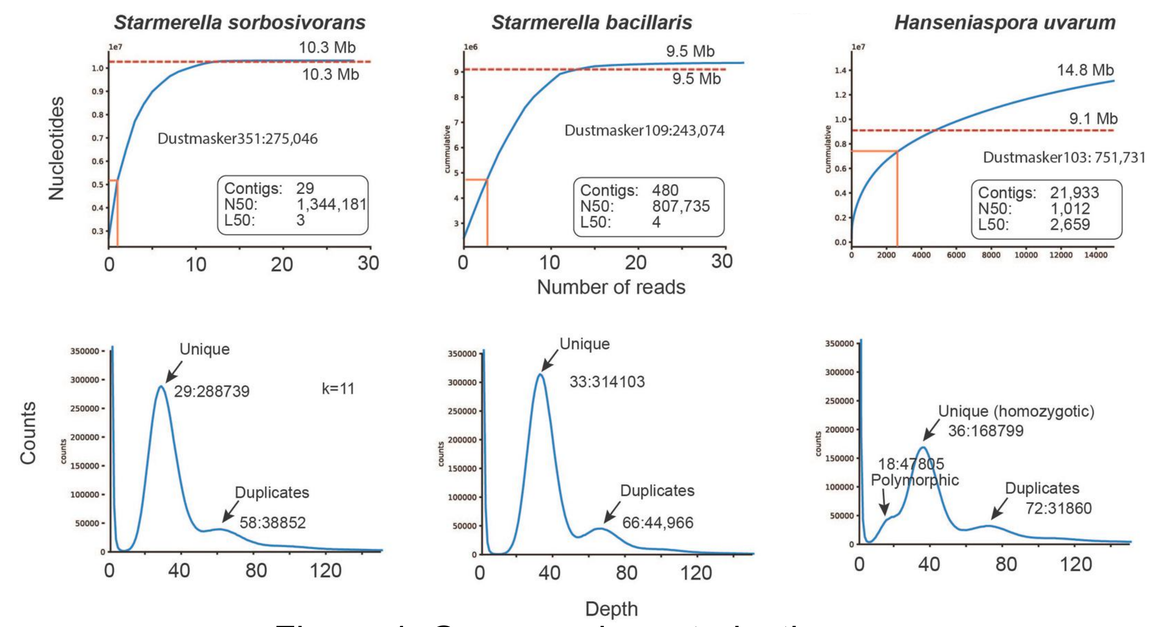

Figure 4. Genome characterization

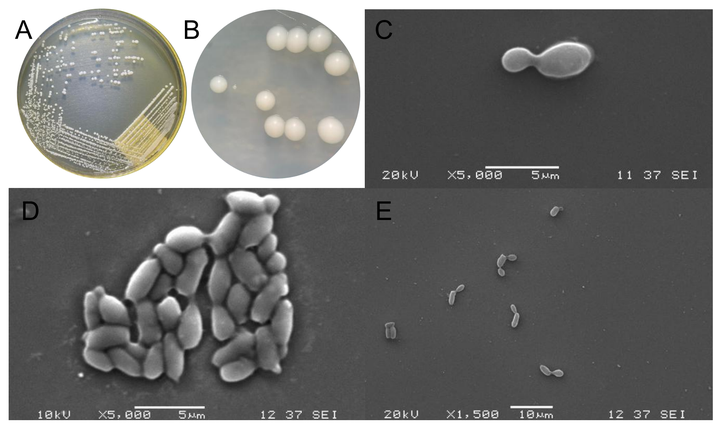

Figure 5. A) *S. bacillaris* B) *S. sorbosivorans* C) *S. sorbosivorans* D) *H. uvarum* E) *S. bacillaris*

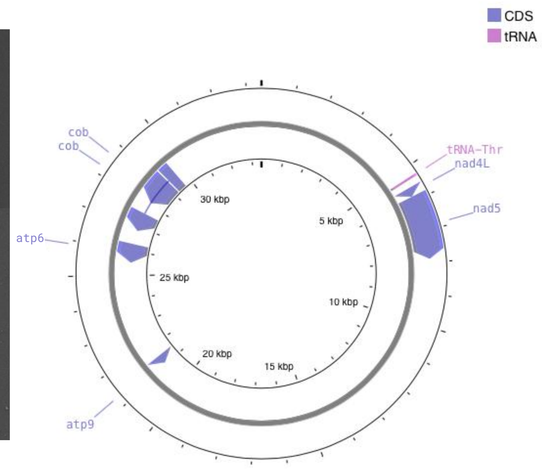

Figure 6. *S. sorbosivorans* mitochondrion

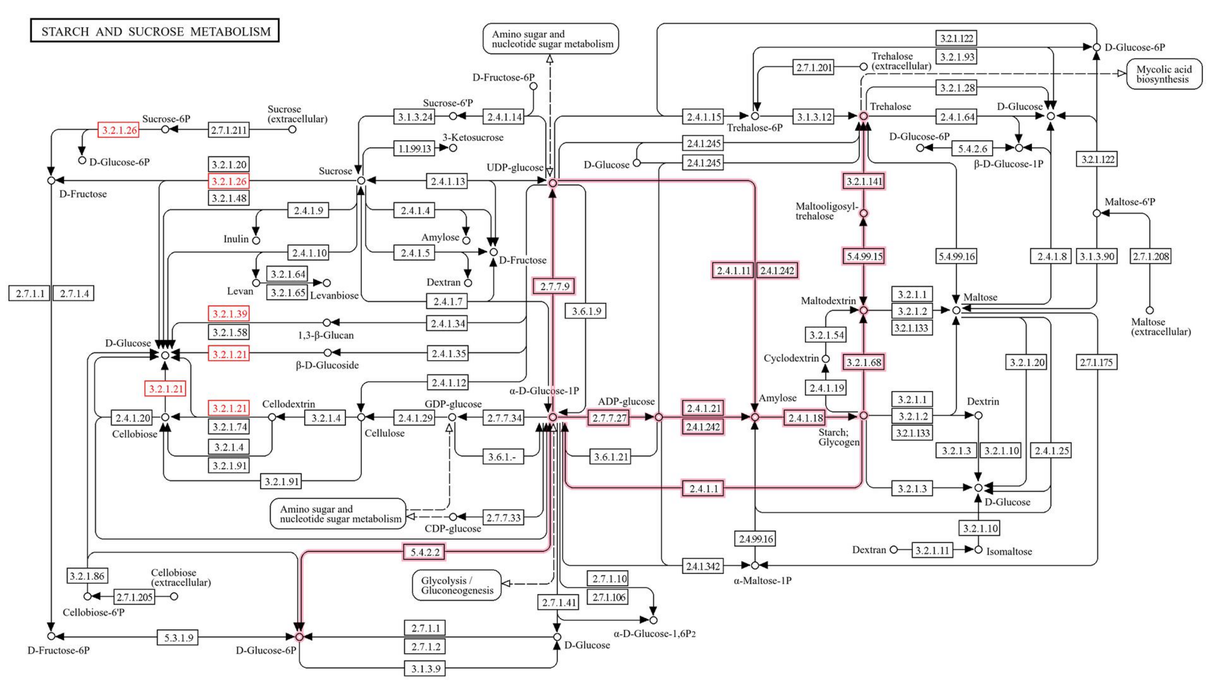

Figure 7. Glucosidases pathway

Conclusions

- The yeasts were identified as *H. uvarum, S. bacillaris and S. sorbosivorans*, it should be noted that these species are currently not well-documented in the literature or in the databases.
- The mitochondrial genome sizes of the strains are as follows: *H. uvarum* is 14 kb, *S. bacillaris* is 21 kb, and *S. sorbosivorans* is 33 kb, and they all contain introns.
- The annotated mitochondrial genome of *S. sorbosivorans* is the first to be documented in the NCBI GenBank, with the accession number PQ436977.
- The metabolism of these yeasts is related to the composition of coffee mucilage.
- The annotation and characterization of the genome provide valuable insights to better understand the behaviour of these strains in the coffee fermentation process.

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